Chapter 3. Evaporitic Deposits

1. Introduction:

Evaporites are deposits of mineral salts. The precipitation of mineral salts occurs when the concentration of dissolved ions in a solution exceeds a given threshold. Numerous ionic species can precipitate and form evaporites, which therefore show a certain petrographic and geochemical diversity.

Salinity is measured from the electrical conductivity of water under specific temperature and pressure conditions.
The average salinity of seawater is **35 g/L**, but it is estimated that the salinity of the Dead Sea may have reached up to **460 g/L**. The density of halite (rock salt) is approximately **2.2**, regardless of depth, which makes salt less dense than most other sedimentary rocks (whose density averages around **2.5**). As a result, salt exhibits particular mechanical properties.

The origin of marine salts is continental erosion. Ions are transported to the oceans by rivers. Since marine salinity has remained constant over geological time, there must be sinks that trap these salts. The nature of these sinks depends on the ion concerned:

- **Calcium** is trapped in carbonates,
- **Potassium** is trapped in clay minerals,
- Hydrothermal processes play a major role in maintaining the salinity balance; for example, **albitization** traps sodium.

Living organisms play only a minor role in maintaining this balance, because most of the salts they assimilate are released at the time of their death. The residence time of salts in seawater is estimated at **100 million years**.

2. Climatic and Geodynamic Context of Modern Evaporitic Formations

a. Petrographic and Geochemical Diversity of Evaporites:

Show samples illustrating the different crystalline forms of **gypsum (sulfate)**, as well as samples of **halite** and **sylvite (KCl)** (chloride minerals).
Illustrate the diversity of the chemical compositions of evaporites using the table "*Examples of mineral species from marine and continental evaporitic deposits*" (Fig.01).

Evaporites marines		Evaporites continentales	
Halite	**$NaCl$**	*Halite*	**$NaCl$**
Sylvite	**KCl**	*Sylvite*	**KCl**
Carnalite	**$KMgCl_3$, $6H_2O$**	*Anhydrite*	**$CaSO_4$**
Kainite	**$KMgClSO_3$, 3H2O**	*Epsomite*	**$MgSO_4$, $7H_2O$**
Anhydrite	**$CaSO_4$**	*Trona*	**$NaCO_3$, $NaHCO_3$, $2H_2O$**
Gypse	**$CaSO_4$, $2H_2O$**	*Mirabilite*	**Na_2SO_4, $10H_2O$**
Polyhalite	**$K_2MgCa_2(SO_4)$**	*Thenardite*	**Na_2SO_4**
Bischofite	**$MgCl_2$, $6H_2O$**	*Bloedite*	**Na_2SO_4, $MgSO_4$, $4H_2O$**
Kieserite	**$MgSO_4$, H_2O**	*Gaylussite*	**Na_2CO3, $CaCO_3$, $5H_2O$**
		Glauberite	**$CaSO_4$, Na2SO_4**

Fig. 01. Some mineral species from marine and continental deposits

b. Paralic settings:

Hypersaline lagoons (environments in direct or indirect connection with the sea).
Example: **Lake Assal, Djibouti** — 80 meters of salt were deposited in less than 3000 years. Currently at an elevation of **–150 m**, this lagoon receives seawater through normal faults that connect it to the Gulf of Aden.
Precipitation: **200 mm/year**
Evaporation: **2 m/year**
Salinity: **350 g/L**

Other environments include **sebkhas** and **playas** (Fig. 2).

Fig. 2. Satellite image (Google Earth) of a Sebkha

c. Climatic and Geodynamic Significance of Evaporites

A comparative analysis of the previous cases shows that:

1. **Continental evaporites display a wide chemical diversity**, which depends on the composition of freshwater entering the basin. This composition, in turn, is controlled by the nature of the source rocks in the drainage basin. A given evaporite mineral species does **not** correspond to a specific climatic or tectonic context.

2. **Evaporitic basins are not restricted to warm environments** (e.g., Uyuni).
   The formation of hypersaline brines requires:
   Evaporation > Input of fresh or marine water,
   that is, an **arid climate**.

3. The basin must be **at least partially isolated** — highlighting the role of tectonics and eustatic variations. External water input must be low (low rainfall and limited river water supply).

4. The evaporation of a water body is rapid on a geological time scale; **evaporite sedimentation rates can be high**, and resulting deposits may reach several hundred meters in thickness (Uyuni today; Germanic Triassic: more than **1000 m** thick in the Aquitaine Basin!). Such deposits imply **significant accommodation space**.

In general, the early stages of **continental rifting** appear to favor evaporite deposition (shallow water + subsidence).
However, is this systematically the case when examining ancient evaporitic formations associated with passive margins?

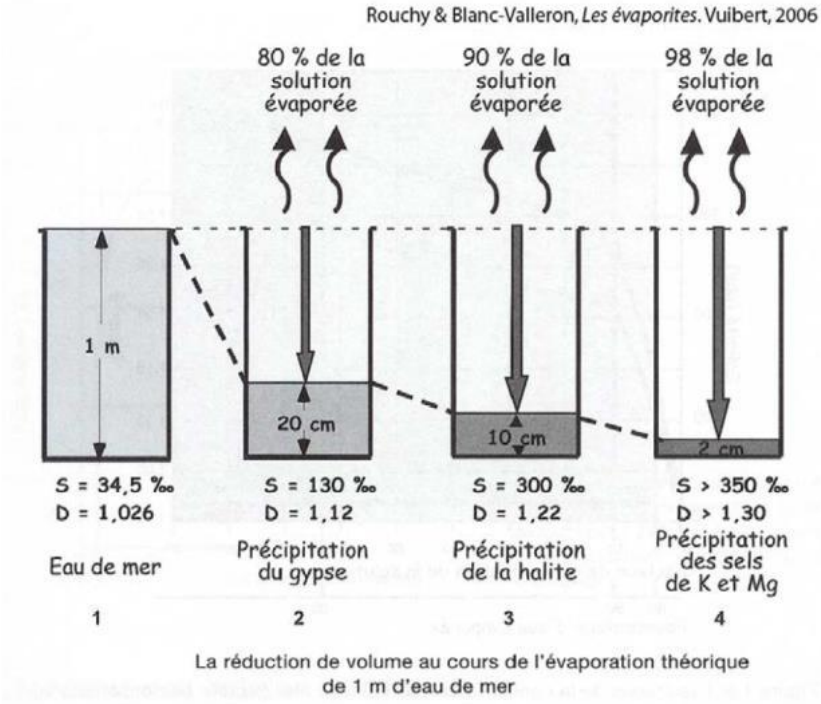
d. Continental settings:
Permanent salt lakes (hypersaline environments with no connection to the sea).
Example: the **Dead Sea**, a pull-apart basin at **–400 m elevation**. Precipitation: **40 mm/year**; very high evaporation.

Hypersaline lakes; example: the **Salar de Uyuni (Bolivia)** (Fig. 3). Located at **3700 m** elevation in the Andean mountain chain, it contains **11 alternations of salt layers and lacustrine deposits** over nearly **150 m thickness**, at temperatures below **8 °C (!)** and with annual precipitation of **200–300 mm**. Evaporation: **1.5 m/year**.
(It contains **one-third of the world's lithium resources**.)

Fig. 3.Photograph of the Salar de Uyuni in Bolivia

e. How are evaporites deposited?

i. Evaporation.



ii. Models of evaporite deposition (Fig. 4):

a. Permanent surface inputs (channels, fractures)

b. Episodic surface inputs (storm surges, spring tides)

c. Subsurface flow (permeable barriers, karstified carbonate barriers, fractured barriers)

d. Hydrostatic differential

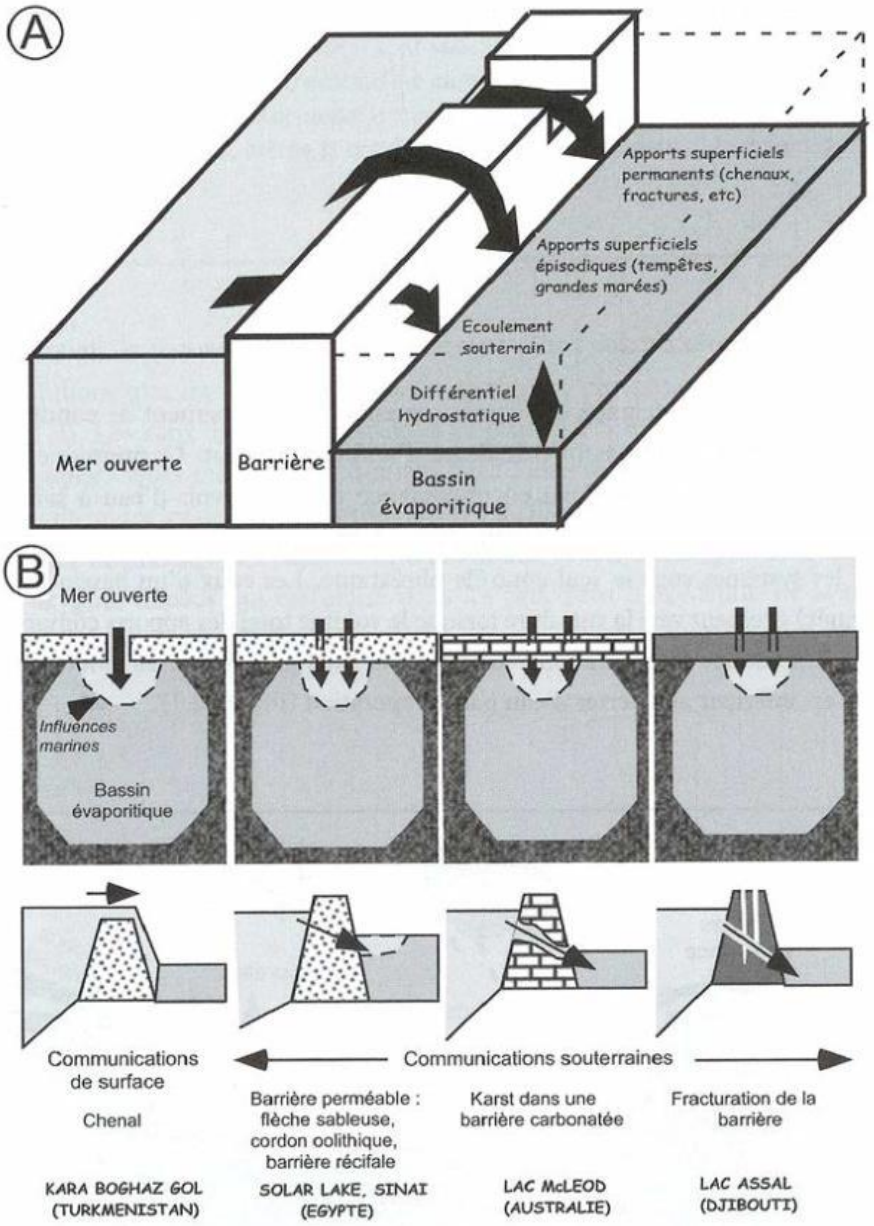


Fig. 4. Evaporite deposition models

3) Climatic and Geodynamic Contexts of Ancient Evaporitic Formations

a. Evaporite deposits associated with epicontinental seas:
Example: **Triassic of the Aquitaine Basin**

- Germanic facies: *Buntsandstein* (red sandstones), *Muschelkalk* (marls and evaporites), *Keuper* (lagoonal evaporites).
- Marine transgressions are linked to the opening of the **Tethys Ocean** and to the increase in ridge volume.

Example: the **Zechstein Sea** (Germany).
This setting follows the "**shallow water, shallow basin**" model, where eustatic variations promote the isolation of a basin.

b. The Messinian Salinity Crisis (5.3 Ma): tectonics-induced basin isolation
The convergence between **Africa and Eurasia** caused the closure of the **Strait of Gibraltar** around **6 Ma**, isolating the Mediterranean from the Atlantic Ocean.
Consequences:

- **≈ 1 million km^3 of evaporites deposited**
- Local evaporite thickness reaching **up to 1500 m**

4) Salt Tectonics: Diapirism

How are salt diapirs, frequently observed in seismic reflection, formed?

For a long time, it was assumed that the salt–sediment interface (with sediments overlying salt) could spontaneously wrinkle due to **Rayleigh–Taylor instability**. However, the strength of the overlying sediments is too high to allow folding and fracturing solely by convective destabilization, as the stresses generated are too weak.

A commonly accepted model is that **diapirism initiates by differential sediment loading**:

- Areas with a thicker sedimentary pile exert stronger pressure on the salt than areas with thinner sediment accumulation.
- Salt, being ductile and mobile, flows to compensate for this load difference, leading to diapir formation (Fig. 5,6).

However:

- Salt diapirs **cannot pierce through their sedimentary cover on their own**.
- This would require localized non-deposition zones to allow salt to rise to the surface — conditions that **do not occur naturally**.

Although variations in sediment thickness are present in all basins, **evaporitic layers do not systematically develop diapirism**.

The origin of diapirism lies elsewhere:
An **extensional regime** must act on the sedimentary cover above the salt to initiate faulting. These faults then allow salt to migrate upward into the sedimentary pile.

Salt diapirism is therefore a **syn-sedimentary process** that deforms sedimentary units differently depending on its growth stage.

Importantly, during diapir development, **excellent petroleum traps are formed**.

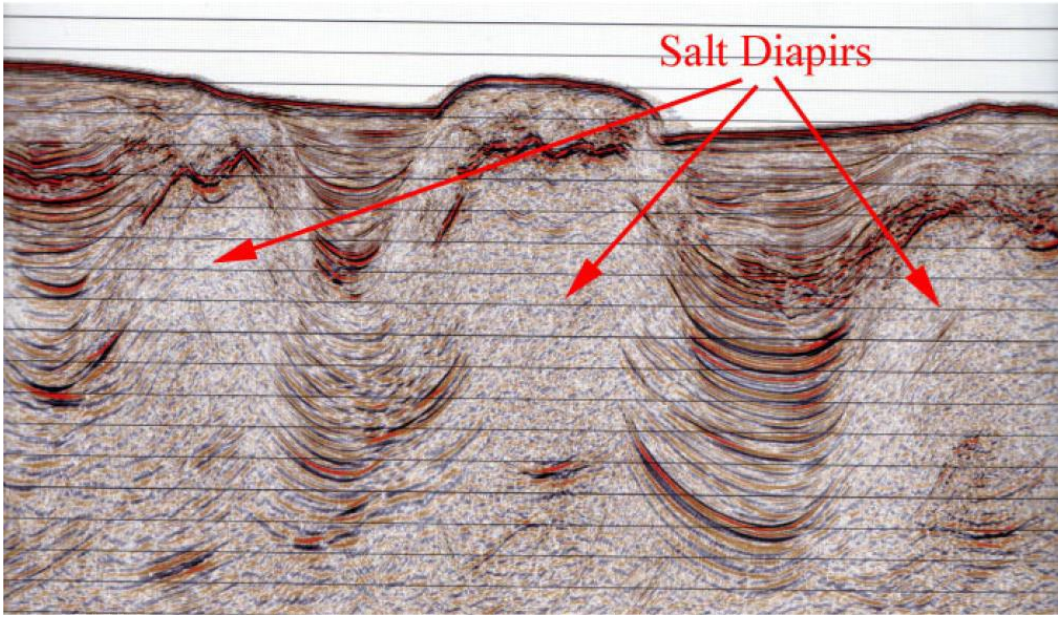


Fig.5. Example of salt diapirs in seismic reflection, observed off the coast of Brazil

Fig.6. Diapir of Ain Ouarka, Western Saharan Atlas (Ain Sefra, wilaya of Naama)

III.2. CONTINENTAL EVAPORITES

In reality, evaporites in continental depressions result from simple **recycling**: salts brought to the surface in an **endorheic drainage basin** are dissolved and transported by water. In the absence of sufficient drainage, these salts are redeposited at the bottom of the depression, where the water evaporates. This is essentially a case of **reworking**.

1. THE SEBKHA ENVIRONMENT

Introduction

i. Definition and Characteristics

Sebkhas are **coastal plains** or slightly depressed surfaces located above the tidal range, or **endorheic basins** with no outlet to the sea. They are encrusted with salts (evaporites) and may grade into **salt marshes or lagoons**, which are temporarily flooded.
Their **intracontinental equivalent** is the *playa*.

A. Definition of evaporites:
Evaporites are **rocks formed by precipitation of alkaline and alkaline-earth salts**. As the name implies, they originate mainly from the **evaporation of saline solutions**, whether from seawater or from saline lakes.

B. Main characteristics:
Some sebkhas are very extensive: **6,000 km^2**, with widths ranging from **80 to 100 km**, and a very low slope of approximately **1/2000** toward the coast.
Groundwater lies very close to the surface. It is **saturated or oversaturated in gypsum and anhydrite**, and sometimes contains **halite and sylvinite**.
The crystals are commonly **lenticular**, or exhibit shapes described as **"spearhead"** or **"lark's foot"**.

ii. Types of Sebkhas

A. Coastal sebkhas:
These are **supratidal** and consist of **marine carbonates** overlain by deposits of **algae and stromatolites**. This environment undergoes intense diagenetic transformations (Fig. 7).

Example: Sebkha of Abu Dhabi.

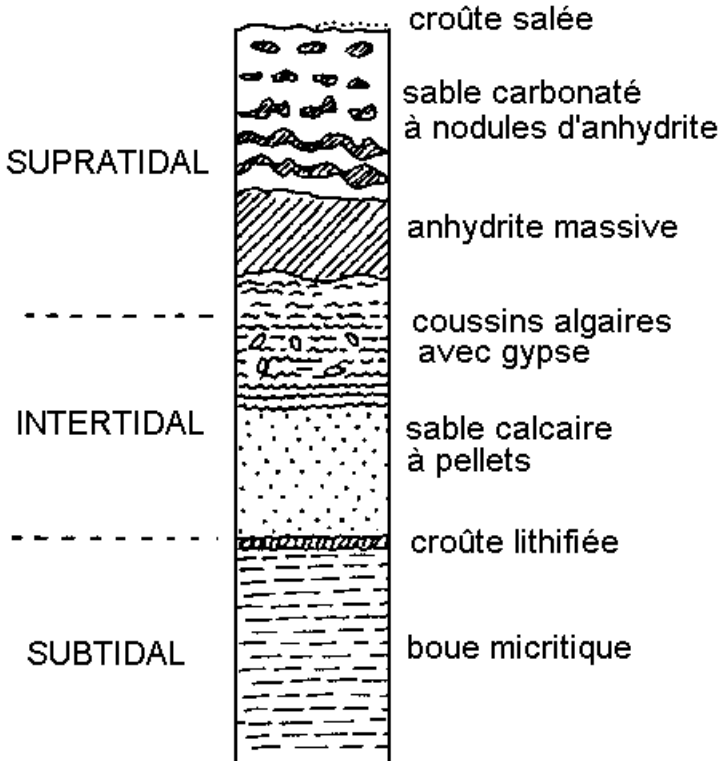


Fig.7. Evaporite sequence in the Abu Dhabi sebkha; the total thickness of the terrains crossed is approximately 02 m.

The sebkha deposits prograde toward the lagoon. In a cross-section of a sebkha, one observes that **supratidal gypsum and anhydrite levels overlie the layers deposited in intertidal and subtidal environments**: the evaporitic sequence is **regressive**.

B. Continental sebkhas

These sebkhas are located landward of the coastal ones and represent **surfaces where sedimentation and aeolian deflation are in equilibrium**, with the depression reaching the level of the groundwater table.

- Sand dunes may develop on these surfaces, along with **alluvial deltas or alluvial fan deposits**.
- **Local carbonate or saline crusts** may form.
- The water table lies **1 to 2 meters below the surface**, and the **capillary fringe stops deflation**.

- Sediments deposited are **quartz grains or limestone grains** eroded from older reliefs.
- Aeolian dune structures are **partially preserved**.

iii. THE EVAPORITE SEQUENCE

Evaporite successions theoretically display a vertical sequence that corresponds to the final stage of the **ideal sedimentary cycle of Goldschmidt** and the **generalized evaporite sequence of Lombard (1956)**. From bottom to top, these series include:

- **Clays and/or bituminous black limestones**
- **Anhydrite ($CaSO_4$)**
- **Rock salt (halite, NaCl)**
- **Potash salts (sylvite, KCl) and magnesium salts (carnallite, $MgCl_2$)**

These deposits occur in **thin laminae**, recording **climatic variations and solar-radiation cycles (Milankovitch cycles).**

iv. DEPOSITIONAL ENVIRONMENTS AND ASSOCIATED LITHOFACIES

Clays and black limestones are associated with **organic matter**. They are deposited in **stagnant, reducing waters** filling depressions. Laterally, these sediments grade into **platform limestones and dolomites**, which may be accompanied by **isolated reefs or reef chains**.

Biological processes play a major role in the deepest parts of the basins. **Bacteria and other microorganisms reduce sulfates into sulfides (H_2S).** Bituminous horizons form and are often associated with an abundance of organisms.

Benthic faunas are absent; only **pelagic or planktonic forms** occur, transported from the shoreline or surface waters.
Organic sedimentation is interrupted by an evaporite phase, with gypsum followed by **potassic and magnesian salts**.

When less saline water enters the basin, a **reverse sequence** of salts is deposited, and the evaporitic series ends with a **clay-rich episode**.

The repetition of evaporite sequences results from **periodic variations in brine concentration**, depending on the balance between **evaporation** and **freshwater or seawater inputs**.
Since these variations are partly climate-controlled, such cycles are interpreted as **climatic cycles or reflecting solar activity** (evaporite laminae). Rhythms of **12–16**

years, 80–90 years, and 180 years have been reported in Jurassic evaporite cycles in New Mexico.

Successive precipitation from a concentrating brine produces the sequence **a–b–c**.
It is often followed by a **c–b–a reverse sequence**, corresponding to the deposition of salts in **descending order of solubility**.

v. EVAPORITE CYCLE (= MAJOR EVAPORATION CYCLE) (Fig. 8)

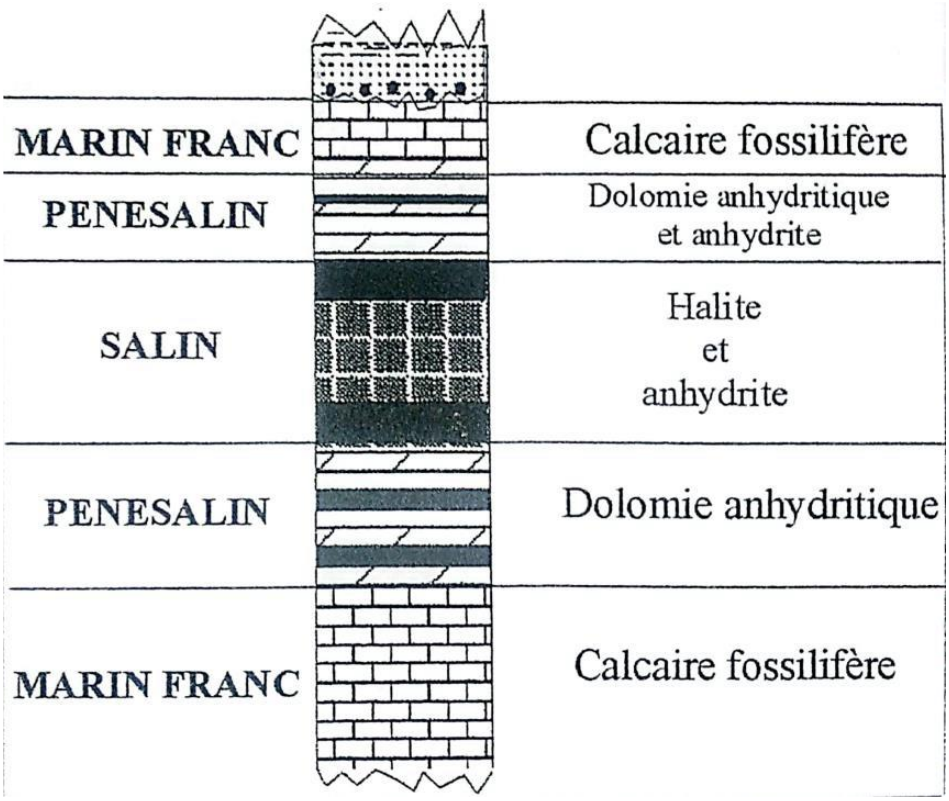


Fig.8. Synthetic representation of a major evaporite cycle

Evaporites have a specific place both in the stratigraphic series in the vertical sense and in the horizontal sense, in other words, in their geographic distribution.

The figure..., shows how evaporites are positioned in the sedimentary cycle: **a marine facies**, represented by a fossiliferous limestone, is followed by a **penealine facies** with **dolomite and anhydrite**. As evaporation intensifies, we move on to **saline facies** with halite and anhydrite. The gradual return to marine facies brings back the facies seen previously in reverse order. The same distribution (Fig.9) of facies is observed **horizontally** according to the **'Law of Walther'**.

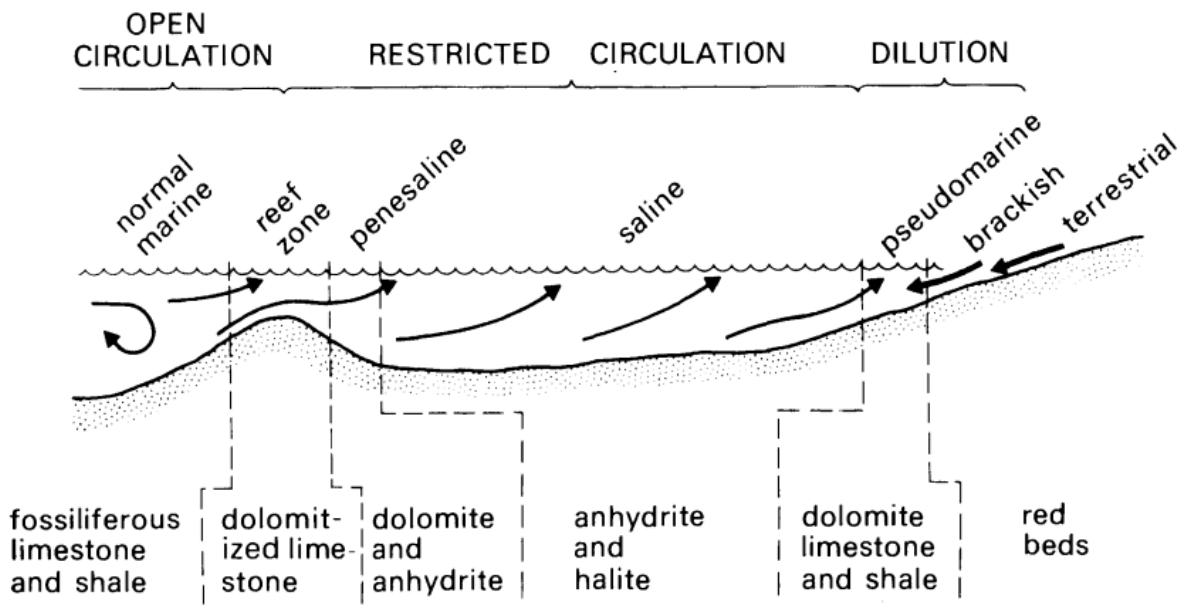


Fig. 9. Profile showing the relationships of topography and surrounding environments in the case of an accumulation of evaporite within a basin.

vi. SOME TYPES OF CLOSED BASINS

A. Dayas (small lakes): These are more or less large depressions of various shapes with a permeable bottom. Water that reaches them does not stagnate; it drains underground.
There are many dayas in the South and in the Tell region. A good example of a daya is found in the Sahel of Oran, between the Arzew Massif and the Saline: this is the Télamine daya; it is an elliptical depression measuring 6 km in length and 1 km in width. The bottom is sandy with marine shales.

B. Playas: This term refers to water bodies with high concentration (salinity) that develop in endorheic basins.

C. Sebkhas: These are typical closed basins, with no outflow either at the bottom or around the edges. They are temporarily flooded by water from rainfall, floods, groundwater resurgence, or even by capillary rise. The bottom is impermeable. Water that reaches them remains stagnant and disappears only through evaporation, after depositing all suspended or dissolved materials. When rich in salt, the latter forms a crust, and during the dry season, the lake is at least partially transformed into a salt flat. Example: the Grande Sebkha of Oran-Aïn Temouchent.

D. Chotts: Chotts are often very large, flat, and salty depressions, rarely containing usable water, but always remaining moist. They occupy the low parts of endorheic

systems in North Africa and the Sahara. They are zones of wind deflation, which leaves witness mounds (Gour).
These are closed basins fed by notable wadis; for example, the Grand Chott Chergui, fed by springs and rivers (gypsum, chloride). Zahrez are typical chotts; their bottom is clayey-sandy. Examples: Chott Rharbi, Chott el Hodna, meaning "plain surrounded by mountains."

2. LACUSTRINE ENVIRONMENTS

(Freshwater or saline lakes)

i. GENERALITIES

The study of lakes is a science called **Limnology**.
The study of lakes depends much more closely on the geological structure of the basin than that of the oceans, which are more uniform.
Climate also greatly influences the deposited material (rainfall regime). Large lakes show some resemblance to the sea: there is a sorting of materials toward the open water. The shores, battered by waves, can include beaches.
When rivers flow into a lake, they build deltas. If the lake level remains constant, which occurs mainly in large lakes, the delta exhibits a simple structure.

ii. DEFINITION

Lakes are natural bodies of freshwater, or (rarely) saline water, located inland, whose extent and depth are variable. These waters do not communicate with the sea and represent an expansion of the hydrographic network.
Lakes are characterized by their:

1. Isolation;
2. Relatively shallow depth;
3. Little or no waves and currents;
4. Predominance of climatic factors.

iii. DIFFERENT TYPES OF LAKES

Lakes have very different origins. From this point of view, they can be classified into two types:

A. External origin:

1. By dissolution
2. Glacial action

B. Internal origin:

1. Tectonic lake: by subsidence or folding
2. Volcanic lake

iv. COMPOSITION OF LACUSTRINE WATERS

Lacustrine waters contain various salts of Ca, Si, Al, Mg, Fe, as well as Cl and S, the amounts of which depend on the nature of the terrains crossed by the lake's tributary waters.

The concentration depends on climate, precipitation, evaporation, and outflows. We can distinguish five categories:

1. **Ahaline waters:** < 0.01 g of salts/l; for example, rainwater or peat bog water. These waters are poor in flora and fauna and are acidic (low pH).
2. **Oligohaline waters:** 0.01 to 0.5 g/l. Most freshwater lakes belong to this type. Most rivers, streams, and drinking waters also fall into this category.
3. **Mesohaline waters:** 0.5 to 15 g/l. These are so-called brackish waters, found in former closed seas (e.g., the Caspian Sea), seas receiving large amounts of freshwater (e.g., the Baltic Sea), or lakes communicating with the sea (e.g., lagoons).
4. **Polyhaline waters:** 15 to 45 g/l. Marine waters belong to this type: Black Sea: 17 g/l; Atlantic Ocean: 35 g/l; Red Sea: 40–45 g/l.
5. **Ultrahaline waters:** > 45 g/l. Most saline lakes belong to this category. The Dead Sea is particularly rich in salts: 350 g/l. These waters are loaded with carbonates, sulfates, and chlorides.
    - Calcium carbonates and sulfates precipitate when the water reaches the Dead Sea.
    - Chlorides remain in solution.
    - Historically, the Dead Sea was much larger than it is today; it concentrated locally through evaporation.
    - It should be noted that Dead Sea waters contain no iodine and little zinc, and thus do not represent the remains of an ancient sea; they are typical of a collapse-type lake.

Remarks:

- A true "mineral" stratification exists in the water. The most saline waters are found at depth. Surface waters, which are oligohaline and purified by biological activity and solar radiation, are often used for drinking water.

- Water color: at great depth, water appears blue. Iron salts can turn it green; however, in contact with calcium and humic compounds, the water becomes clearer.

v. LACUSTRINE SEDIMENTATION

A. Detrital Sedimentation:

1. Nature:
Detrital lacustrine sedimentation consists of pebbles, sands, and muds.

2. Modalities:

a. Deltaic Sedimentation:
Deposits are well developed at the lake margins and at river mouths, forming a **lacustrine delta**. Sedimentation is characterized by an alternation of coarse and fine sediments, producing **cross-bedding** that reflects river floods.

- If the lake level remains constant, the delta shows a simple structure:
  - Horizontal layers at the top,
  - Oblique layers at the delta front, steeply dipping toward the lake,
  - Horizontal bottom layers beneath.
- As the delta progresses, front layers overlie previously deposited bottom layers.
- If the lake level fluctuates, the delta structure becomes more complex.

b. Lacustrine Varves (from the Swedish *varv* = layer):
These are thin layers that alternate between light and dark.

- Dark layers form in winter (organic matter and iron sulfide),
- Light layers form in summer and are more calcareous.

Varves provide a way to measure **sedimentation rates**.

- In some lakes, abundant organic matter (**sapropels**: mud containing a high proportion of organic material) produced by phytoplankton can lead to **bituminous rocks** of the pyrobituminous type. This means the bitumen is extractable only by **pyrogenation** (a heat-induced chemical reaction) and is not soluble in organic solvents, unlike petroleum products.

Sediment deposits vary depending on climate and the nature of the catchment area:

- In **Scotland**, sands, clays, brown muds, and sometimes diatomaceous muds are deposited.

- In the **Alps**, where limestone is abundant, sedimentation mainly produces lacustrine chalk, occasionally algal deposits, and very rarely calcareous oolites.

Fig.10. Varve deposition

c. Glacio-lacustrine Varves:

This is **periodic sedimentation** in glacial lake formations. It consists of a very fine alternation of dark clayey layers and lighter, more quartz-rich layers.

- When the lake is frozen, the water is calm, allowing fine clay particles to settle, forming **dark layers** colored by organic matter.
- During thawing, water and detrital material are brought in, forming **light sandy layers**.

Sometimes, **sub-lacustrine sliding phenomena** occur within varves, producing **convoluted layers** (*convolute laminations*). The material, with a consistency like toothpaste, flowed over a very gentle slope.

Since varves are a **seasonal phenomenon**, they provide a means of **directly dating a deposit by counting layers**. Determinations of this kind in Switzerland have measured deposit durations on the order of **20,000 years**.

Fig.11. Convolute laminations

B. Chemical and Biochemical Sedimentation

1. Calcareous and Clayey Sedimentation:

a. Pure Lacustrine Limestones:
Their formation is linked to the evolution of the following equilibrium:

$$2HCO^{3-}+Ca^{++}\longrightarrow CaCO_3+CO_2+H_2O$$

An increase in temperature or water agitation facilitates CO_2 release.

- CO_2 absorption by aquatic green plants in light leads to the precipitation of $CaCO_3$.

Near the shores, this process forms **calcareous tufa**, with more or less spongy textures, and **travertine**, which is finer-grained and more compact.

b. Lacustrine Chalks:
These mainly form at the center of lakes by the accumulation of surface debris from living organisms, combined with varying amounts of chemically precipitated material or very fine clay particles.

- In cross-section, chalk exhibits a **striped appearance**, with alternating light and dark zones, each approximately 0.1 mm thick.
- This pattern reflects **seasonal variations**, resulting in a biological cycle with two main phases:

Winter:

- Chlorophyll activity slows, and most phytoplankton dies.

- Organic matter accumulates and is decomposed by putrefactive bacteria (cold slows putrefaction).
- Harmful products such as NH_3 and H_2S may appear, potentially killing remaining plankton.
- Oxygen levels drop, CO_2 accumulates, and pH is neutral to acidic.
- Clay particles, enriched with organic matter, flocculate in colloidal suspension, forming **dark layers**.

Summer:

- Chlorophyll activity resumes, releasing significant O_2, which both aerates the water and oxidizes ammonia, hydrogen sulfide, and organic phosphorus into nitrites, nitrates, sulfates, and phosphates usable by plants.
- CO_2 consumption increases, taken from bicarbonates, causing precipitation of Ca and Mg carbonates from solution.
- Insoluble $CaCO_3$ forms, making the water appear milky, and eventually deposits as **white layers**.

Thus, the white chalk deposited in the warm season is the result of true **biogenic decalcification**, forming the clear zones observed in the sediment.

3. MARINE EVAPORITES

A.1. Shallow Basins:

There are two models of shallow basins:

a. Closed basins:

In these basins, facies are organized concentrically, like an “eye,” with the **least soluble salts** (carbonates) at the periphery and the **most soluble salts** (potash salts) at the center.
Such an arrangement would produce a deposit in which the percentages of the various salts are close to those obtained by **experimental evaporation of seawater**.

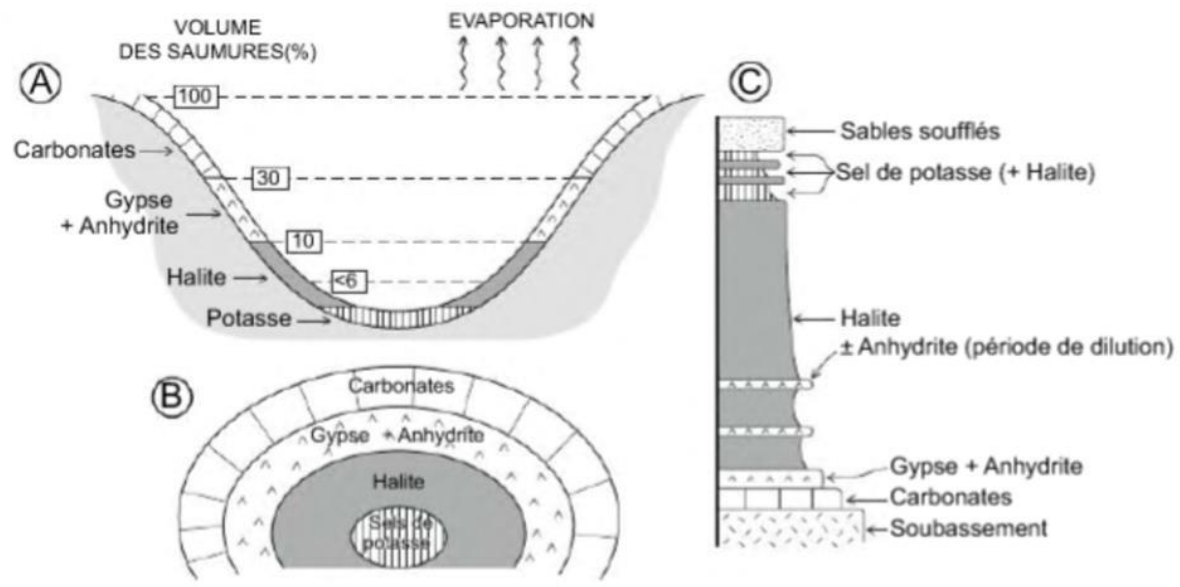


Fig.12. Evaporites of a closed marine basin (after Einsele, 1992)

A: Evolution of sedimentation as a function of the degree of evaporation

B: Distribution of facies on a map

C: Typical evaporitic sequence

b. Semi-isolated basins:
These basins maintain a more or less significant connection with the open sea (threshold basins). Facies are then organized according to a polarity centered on the channel connecting the basin to the sea:

- **Least soluble salts** are near the channel,
- **Most soluble salts** accumulate at the opposite end, at the bottom of the lagoon (arrangement like a “drop of water”).

In the evaporitic sequence, the **carbonate and calcium sulfate stages** are more developed than the **halite and potash salt stages**, which are often not reached at all.

A.2. Deep Basins:
These can be divided into two types:

1. Deep-water basin models:
Due to restricted communication (eustatic changes, tectonics), deep circulation in a deep basin (on continental or oceanic crust) is replaced by **surface circulation**.

- Phytoplanktonic production remains significant near the surface, while stagnation occurs in the deeper waters, creating **euxinic conditions** (bituminous marl deposits).
- Surface waters, becoming denser and saltier due to evaporation, begin to sink into the deepest parts of the basin, while **gypsum and halite** start to precipitate from the surface waters.

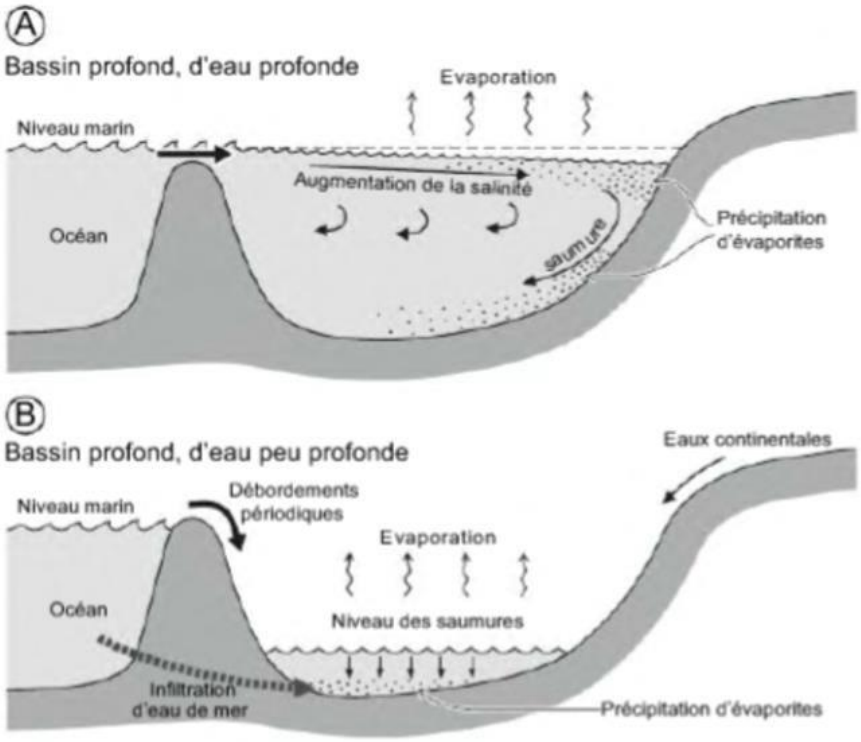


Fig.13. Deep evaporite basins

1. Deep and Shallow Water Basin Models:

In this case, a **"structurally deep" basin** becomes completely isolated from the ocean, leading to **on-site drying** and the development of **evaporitic facies** in shallow water or sebkha environments.

4. CONCLUSIONS

1. It is difficult to directly link evaporites to a specific climate or geodynamic context. This study has shown the diversity of environments in which salts are deposited. Some common factors emerge: to form evaporite deposits, there

must be a **combination of arid climate**, a **shallow water body**, and **preferential isolation from any additional water input**.

2. The study of **lacustrine sedimentation** provides a means to measure deposition rates and establish chronological sequences.
3. Offshore, phytoplankton flourishes; the resulting organic matter, mixed with mud, forms **sapropels**—the source material for **bituminous sediments** and potentially some petroleum. In lakes rich in silica (near tuffs or eruptive rocks), **diatoms** can proliferate. Their frustules accumulate on the bottom, forming **diatomites**.
4. In **marginal lacustrine zones**, peat bogs may develop, leading to deposits such as **peat** and **wooded marshes**, which can form **lignites and coal seams**.
5. **Salars** (salt flats) contain significant quantities of **lithium**.
6. **Diapirs** (salt domes) favor the formation of **petroleum traps** in the sedimentary formations they penetrate.
7. Salt is primarily used for **road de-icing** during icy conditions.